

MUHAMMAD YASEEN

Senior Full-Stack AI Engineer

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PROFESSIONAL SUMMARY

Senior Full-Stack AI Engineer with **3+ years of experience architecting and deploying 20+ production AI systems** across computer vision, large language models, generative AI, and IoT domains. Currently leading AI/ML initiatives at One Network (FWO), designing intelligent microservices for national-scale network optimization, infrastructure monitoring, and predictive analytics. Previously built fraud detection systems processing 20M+ daily events for the Amazon advertising ecosystem at EmporionSoft, and pioneered automated multimedia-to-documentation ML frameworks at Evolve Innovative Solutions. Published peer-reviewed research on multi-disease prediction. Proficient in end-to-end AI product lifecycle — from research and data engineering to cloud deployment and continuous monitoring.

CORE COMPETENCIES

Core AI & Machine Learning: PyTorch, TensorFlow / Keras, Scikit-learn, Deep Neural Networks, Transformers & Attention Mechanisms, Siamese Networks & Metric Learning, MLOps (MLflow, W&B, SageMaker)

Computer Vision: YOLOv5/v8/v10, OpenCV, Object Detection & Tracking, Image Segmentation, Feature Extraction & Matching, Video Analytics

LLM & Generative AI: LangChain / LlamaIndex, RAG Pipelines & Vector DBs, GPT / Claude / Gemini Integration, Prompt Engineering & Fine-Tuning, Stable Diffusion & ControlNet, LoRA & DreamBooth Fine-Tuning, Whisper STT / ElevenLabs TTS

Backend & Infrastructure: Python (Expert), FastAPI / Flask, PostgreSQL / Redis, Docker & Kubernetes, Apache Spark, AWS (SageMaker, EC2, S3, Lambda), CI/CD Pipelines (GitHub Actions)

Frontend & Mobile: Next.js / React, TypeScript, React Native, Tailwind CSS, State Management (Zustand, Redux)

IoT & Real-Time Systems: RFID / NFC Integration, MQTT & Real-Time Streams, Sensor Fusion, Embedded Systems Integration

PROFESSIONAL EXPERIENCE

Senior AI Engineer

Mar 2025 – Present

One Network – FWO | Islamabad, Pakistan

- Architected a scalable AI microservice ecosystem for real-time network traffic optimization, achieving 99.7% uptime and sub-100ms inference latency across 50+ edge nodes.
- Engineered generative AI pipelines producing high-fidelity synthetic telemetry data, reducing sensor dependency costs by 40% while improving anomaly detection recall from 82% to 96%.
- Designed computer vision surveillance systems utilizing YOLOv8 and custom CNN architectures to automate infrastructure threat detection, cutting manual inspection overhead by 65%.
- Integrated LLM-powered conversational agents into customer-facing support systems, reducing ticket resolution time by 58% through intelligent triage and contextual response generation.

AI Engineer

Jan 2024 – Feb 2025

EmporionSoft Pvt Ltd | Lahore, Pakistan

- Architected a DNN-based crawler detection framework processing 20M+ daily traffic events, increasing robotic coverage by 11.9% and Impression IVR by 4.9% — translating to \$2.3M in annualized savings for top-tier advertisers.
- Designed multi-stage ML pipelines combining behavioral fingerprinting, request pattern analysis, and real-time ensemble scoring to identify and block sophisticated robotic fraud with 97.3% precision.
- Conducted rigorous statistical root-cause analysis on data quality degradation, implementing automated validation layers that reduced false positive rates by 34% and restored model confidence scores.
- Developed an adversarial robustness training protocol that hardened detection models against evolving bot signatures, maintaining 95%+ detection efficacy across 3 major botnet evolution cycles.

Machine Learning Engineer

Sep 2023 – Dec 2023

Evolve Innovative Solutions | Lahore, Pakistan

- Engineered a novel ML framework combining video understanding (action segmentation, object detection) with NLG models to auto-generate step-by-step technical documents from raw procedural videos.
- Achieved BLEU score of 0.74 and ROUGE-L of 0.81 on generated documentation quality — outperforming existing benchmarks by 18% through custom attention mechanisms and domain-specific fine-tuning.
- Validated framework robustness through empirical evaluation across 6 diverse procedural domains (manufacturing, lab protocols, IT operations, medical procedures, cooking, assembly), demonstrating cross-domain generalization.

KEY PROJECTS

Vehicle Visual Similarity & Matching Engine

Deep learning engine that matches vehicles across camera feeds using visual embeddings for toll fraud detection and smart city enforcement.

Impact: Reduced manual vehicle verification time by 82% and identified 14,300+ toll evasion incidents in the first quarter of production deployment.

Tech: Python, PyTorch, FastAPI, Siamese Networks, Vector Embeddings, PostgreSQL, Docker, Redis

AI Call Center Voice Bot

End-to-end voice AI pipeline — from curating 20K call recordings to deploying a live STT → classification → LLM → TTS voice bot.

Impact: Autonomously resolved 73% of inbound customer inquiries without human intervention, reducing support staffing costs by 61%.

Tech: Python, Whisper STT, LLMs (GPT-4, Claude), ElevenLabs TTS, BERT, FastAPI, Twilio, Redis

Smart Cities AI Monitoring Platform

City-wide real-time AI surveillance platform integrating camera feeds and IoT sensors for anomaly detection, traffic monitoring, and security incident response.

Impact: *Reduced incident response time by 64% and enabled proactive intervention for 1,200+ traffic anomalies per day across monitored zones.*

Tech: Python, YOLOv8, React Dashboard, OpenCV, MQTT, Apache Kafka, PostgreSQL, Docker Swarm

HRMS Role-Aware AI Chatbot

Enterprise-grade HR chatbot with granular RBAC — dynamically scopes LLM responses based on the authenticated user's organizational clearance level.

Impact: *Eliminated HR ticket backlog by responding to 92% of employee queries autonomously while maintaining zero privilege-escalation incidents in security audits.*

Tech: Next.js, Python FastAPI, LangChain, ChromaDB, GPT-4, RBAC Middleware, PostgreSQL, JWT Auth

Toll Plaza MTag Automation System

Fully automated, contactless toll collection system using RFID readers at plaza entry points with real-time fee deduction and barrier control.

Impact: *Reduced average vehicle processing time from 45 seconds (manual) to under 0.5 seconds (automated), eliminating plaza congestion entirely.*

Tech: Python, FastAPI, React Dashboard, RFID / IoT, PostgreSQL, Payment API Gateway, MQTT, Redis

Imagine Short — AI Photo & Video Generator

Generative AI creative studio for mobile — text-to-image and text-to-video with style presets, aspect ratio controls, and social sharing integration.

Impact: *Achieved 50K+ downloads in first 3 months with 4.6 ★ rating and 32% week-1 retention rate.*

Tech: React Native, Stable Diffusion API, Video Gen APIs, Cloud GPU (RunPod), Redux Toolkit, Firebase, RevenueCat

EDUCATION

Bachelor of Science in Computer Science

3.73 / 4.00

University of Malakand | Khyber Pakhtunkhwa, Pakistan | 2019-09 – 2023-08

Research Paper: *Multi-Disease Prediction Using Machine Learning and Deep Learning Models* — DOI: [10.71146/kjmr176](https://doi.org/10.71146/kjmr176)

CERTIFICATIONS

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- Machine Learning Specialization — Coursera
 - Foundations: Data, Data, Everywhere — Coursera
 - Image and Video Processing: From Mars to Hollywood with a Stop at the Hospital — Coursera
 - IBM AI Engineering Professional Certificate — Coursera
 - Introduction to Data Science in Python — Coursera

LANGUAGES

English — Professional working proficiency

Urdu — Native